

Sustainability aware asset management

Lecture 7: Portfolio Construction under Climate Risks

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Objectives of the Lecture

Several tools can be used in the fight against climate change.

- Regulation is an obvious tool (polluter pays)

However, governments have difficulties to coordinate (see COP)

- Monetary institutions (central banks) can play a role (financial regulation)
- What about finance institutions (banks, asset managers)?

Two main arguments (double materiality)

- Investing in polluting firms has an impact on the planet (fundamental risk)
- Investing in polluting firms is a source of risk for investors (market risk)

Objectives of the Lecture – Readings

Cheng G., Jondeau E., and B. Mojon (2023), *Building Portfolios of Sovereign Securities with Decreasing Carbon Footprints*,

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4207316

Fahlenbrach, R., and E. Jondeau (2021), *Greening the Swiss National Bank's Portfolio*,

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3906654

Jondeau E., B. Mojon, and L.A. Pereira da Silva (2021), *Building Benchmarks Portfolios with Decreasing Carbon Footprints*, SwissFinanceInstituteResearchPaperNo.21-91

Objectives of the Lecture

- **Regulation of Climate Risks**
- Assessing the Climate Performance of a Portfolio
- Building a Decarbonized Portfolio

United Nations Climate Change Conference: Conference of the Parties (COP)

- COP 1: Berlin (1995)
- COP 3: Kyoto (1997) → Kyoto Protocol
- COP 21: Paris (2015) → Paris Agreement
- COP 26: Glasgow (2021)

The **Kyoto Protocol** is an international treaty that commits state parties to reduce GHG emissions, based on the scientific consensus that:

- (1) Global warming is occurring and
- (2) It is likely that human-made CO₂ emissions are driving it

The US did not ratify (dropped out in 2001), and Canada withdrew from the Protocol in 2012.

The **Paris Agreement** is an international treaty with the following goals:

- Keep a global temperature rise this century well below 2°C above pre-industrial levels
- Pursue efforts to limit the temperature increase to 1.5°C
- Increase the ability of countries to deal with the impacts of climate change
- Make finance flows consistent with low GHG emissions and climate-resilient pathways

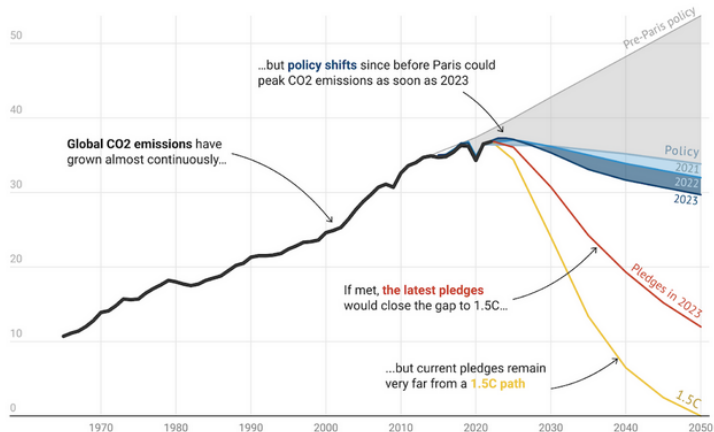
Paris Agreement: where we are?

- 194 states have signed the Agreement (representing about 80% of GHG emissions)
- The US have recently rejoined the Agreement in 2021 (under Biden administration)
- Iran and Turkey have not signed the Agreement

Regulation: United Nations

Global CO2 emissions could peak as soon as 2023, IEA reveals

Global energy-related CO2 emissions, billion tonnes



Source: [https:](https://www.carbonbrief.org/analysis-global-co2-emissions-could-peak-as-soon-as-2023-iea-data-reveals/)

[//www.carbonbrief.org/analysis-global-co2-emissions-could-peak-as-soon-as-2023-iea-data-reveals/](https://www.carbonbrief.org/analysis-global-co2-emissions-could-peak-as-soon-as-2023-iea-data-reveals/)

“We don’t need an army of actuaries to tell us that the catastrophic impacts of climate change will be felt beyond the traditional horizons of most actors - imposing a cost on future generations that the current generation has no direct incentive to fix.” (Mark Carney, 2015)

Some central banks, regulators and supervisors are already taking steps towards integrating climate-related risks into supervisory practices

Network for Greening the Financial System (NGFS) (<https://www.ngfs.net/>):

Network of 83 central banks and financial supervisors created in Dec 2017

The NGFS recognized that “climate-related risks are a source of financial risk. It is therefore within the mandates of central banks and supervisors to ensure the financial system is resilient to these risks” (NGFS, 2018, p. 3).

Bank for International Settlements (BIS) (<https://www.bis.org/>): International financial institution owned by central banks that “fosters international monetary and financial cooperation and serves as a bank for central banks”. The BIS hosts the Secretariat of the Basel Committee on Banking Supervision (BCBS) and has played a central role in establishing the Basel Capital framework.

Climate-related risk drivers and their transmission channels (2021)

Climate-related financial risks – measurement methodologies (2021)

Task Force on Climate Related Financial Disclosures (TCFD)

(<https://www.fsb-tcfd.org/>): Established in December 2015 by the G20 Financial Stability Board. Its objective is to improve and increase reporting of climate-related financial information.

Recommendations of the Task Force on Climate-related Financial Disclosures (2017)

One Planet Summit: www.oneplanetsummit.fr

One Planet Sovereign Wealth Funds (OPSWF)

Funding members: Abu Dhabi Investment Authority (ADIA), Kuwait Investment Authority (KIA), NZ Superannuation Fund (NZSF), Public Investment Fund (PIF), Qatar Investment Authority (QIA)

One Planet Asset Managers

Funding members: Amundi AM, BlackRock, BNP PAM, GSAM, HSBC Global AM, Natixis IM, Northern Trust AM, SSGA

One Planet Private Equity Funds

Members: Ardian, Carlyle Group, Global Infrastructure Partners, Macquarie Infrastructure and Real Assets (MIRA), SoftBank IA

Investors Alliances

Several initiatives by large asset managers around temperature alignment

Climate Action 100+: an investor initiative to ensure the world's largest corporate GHG emitters take necessary action on climate change. In particular, the Action objective states: “[...] reduce GHG emissions across the value chain, consistent with the Paris Agreement goal of limiting global average temperature increase to well below 2°C”

Institutional Investors Group on Climate Change: an alliance of pension funds and asset managers, with 33 trillion euros of combined AUM, that launched in May 2019 the Paris-Aligned Investment Initiative (PAII) to help investors understand what alignment means in practice as well as identify and review methodologies and approaches. The PAII is producing a framework to support asset owners and managers to align their portfolios and strategies to achieving the goals of the Paris.

Net-Zero Asset Owner alliance: an international group of institutional investors (with over \$4.6 trillion AUM) delivering on a bold commitment to transition their investment portfolios to net-zero GHG emissions by 2050.

Carbon Pricing

Polluter pays principle

- A carbon price is a cost applied to carbon pollution to encourage polluters to reduce the amount of GHG they emit into the atmosphere
- Negative externality

Two instruments of carbon pricing

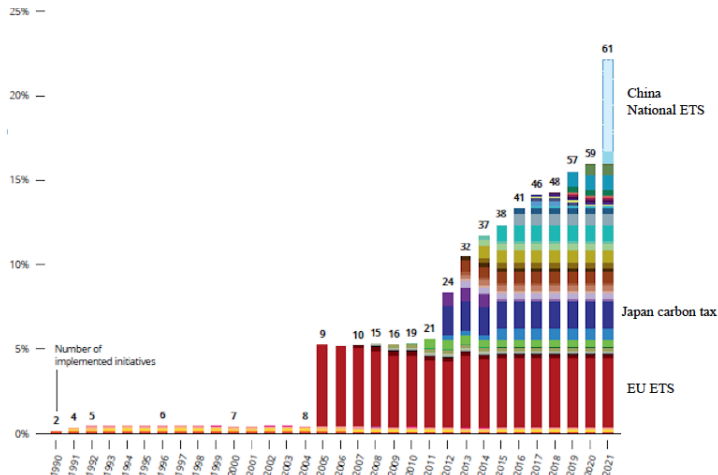
- **Carbon tax** directly sets a price on carbon by defining a tax rate on GHG emissions or – more commonly – on the carbon content of fossil fuels
- **Emissions trading scheme** (ETS) (also called Cap-and-trade system) caps the total level of GHG emissions and allows those industries with low emissions to sell their extra allowances to larger emitters.

Carbon tax: the price is pre-defined, but the emission reduction outcome is not.

Some 40 countries and more than 20 cities, states and provinces already use carbon pricing mechanisms

Carbon Pricing

Share of global emissions covered by carbon pricing initiatives (ETS and carbon tax)



Source: <https://www.carbontax.org/where-carbon-is-taxed-overview/>

Carbon Pricing: ETS

Trading Economics Carbon Emissions Allowances Prices are sourced from the EU ETS, the world's largest cap and trade GHG emissions market (euro/tCO₂)



Source: <https://tradingeconomics.com/commodity/carbon>

Carbon Pricing: Tax

Prices in implemented carbon pricing initiatives (USD)

Country	2018	2019	2020	2021	2022
Sweden	139.11	126.78	133.26	137.24	129.89
Switzerland	100.90	96.46	104.65	101.47	129.86
Norway	64.29	59.22	57.14	69.33	87.61
Finland	76.87	69.66	72.24	72.83	85.10
France	55.30	50.11	6.98	52.39	49.29
Ireland	24.80	22.47	30.30	39.35	45.31
Iceland	35.71	31.34	30.01	34.83	34.25
Denmark	28.82	26.39	27.70	28.14	26.62
Portugal	8.49	14.31	27.52	28.19	26.44
United Kingdom	25.46	23.59	23.23	24.80	23.65
Slovenia	21.45	16.44	20.16	20.32	19.12
Spain	24.8	16.85	17.48	17.62	16.58
Latvia	5.58	5.06	10.49	14.10	16.58
Japan	2.74	2.60	2.76	2.61	2.36

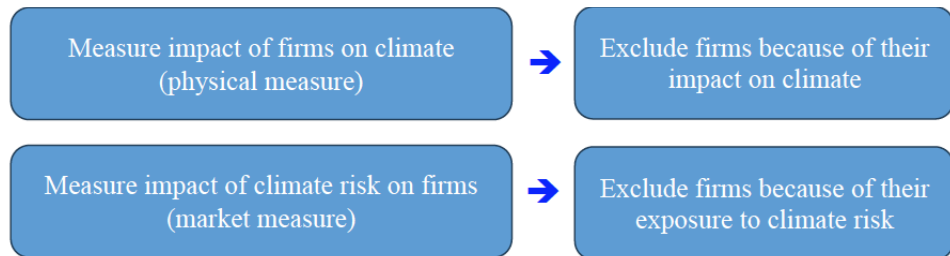
Source: World Bank Carbon Pricing Dashboard, <https://carbonpricingdashboard.worldbank.org/>

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- Building a Decarbonized Portfolio

Physical versus Market Measures

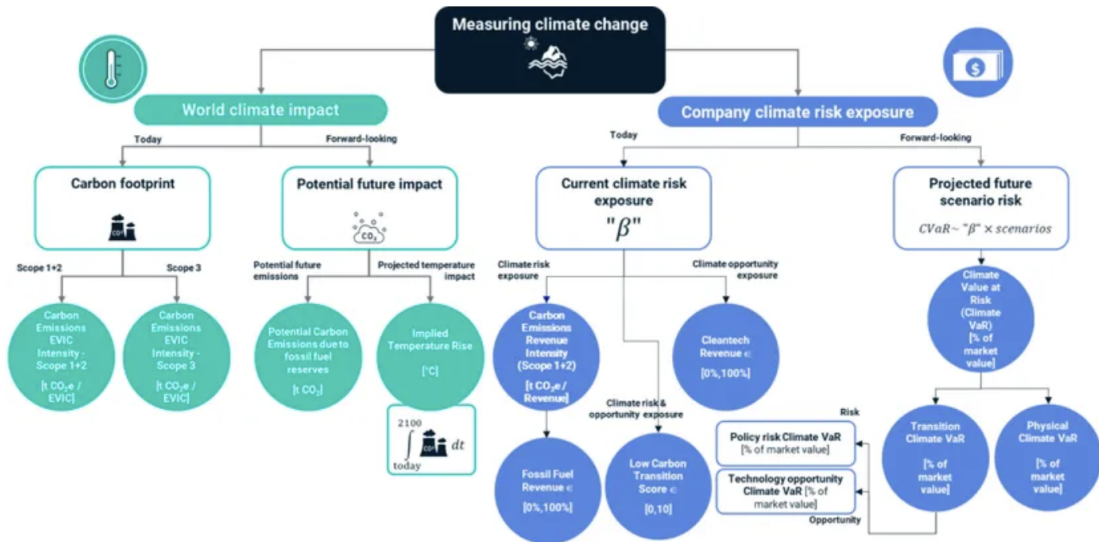
There are two main topics related to **decarbonization** due to double materiality



Next questions:

- Portfolio decarbonization based on current carbon emissions (physical measure)
- Carbon factors (next lecture)
- Portfolio decarbonization based on carbon beta (market measure) (next lecture)

Physical versus Market Measures



<https://www.msci.com/www/blog-posts/how-to-evaluate-climate-metrics/03644475407>

Several ways to measure carbon impact of a portfolio:

- ① Carbon footprint
- ② Proportion invested in “green” and “brown” sectors
- ③ Avoided emissions
- ④ Green-brown metrics
- ⑤ Benchmark

Carbon Intensity Measures

It is natural to normalize carbon emissions relative to a metric of size, quantity or value. Firms in the same industry usually use a physical intensity, financial institutions use a monetary intensity for their portfolio.

- **Physical intensity** measures the carbon emissions of a product or a service relative to the quantity produced: (in kgCO₂e per unit)

Product	Category	Cradle-to-gate	Cradle-to-grave
Computer	Laptop	156	169
	Desktop	169	189
	High performance	295	394
Balloon	Football	3.4	5.1
	Basket-ball	3.6	5.9

Cradle-to-gate: from the production to the store

Cradle-to-grave: entire life of the product

Source: Lhotellier, Less, Bossanne, and Pesnel (2017), ADEME report

- **Monetary intensity** measures the carbon emissions relative to some monetary value

Monetary Carbon Intensity Measures

To compare carbon emissions between firms of different size, normalization is required. A natural indicator is the carbon emissions relative to the revenues (sales) of the firms:

$$CI_{i,t} = \frac{E_{i,t}}{Rev_{i,t}}$$

(1) Economic carbon intensity: Volume of carbon emissions (in tons of CO₂ equivalent) per million dollars of revenue (tCO₂e/mln \$). It measures the carbon efficiency of the portfolio:

$$CI_{i,t}^{(p)} = \frac{\sum_{i=1}^{N_t} o_{i,t}^{(p)} E_{i,t}}{\sum_{i=1}^{N_t} o_{i,t}^{(p)} Rev_{i,t}}$$

- $E_{i,t}$ is the amount of carbon emitted by firm i in year t
- $Rev_{i,t}$ is the revenue generated by the firm
- $o_{i,t}^{(p)} = V_{i,t}^{(p)} / Cap_{i,t}$ is the fraction of the equity of the firm held in the portfolio
- $V_{i,t}^{(p)}$ is the dollar value invested in firm i
- $Cap_{i,t}$ is the market capitalization of firm i in year t

Monetary Carbon Intensity Measures

The carbon intensity can be rewritten as

$$CI_t^{(p)} = \frac{\sum_{i=1}^{N_t} o_{i,t}^{(p)} E_{i,t}}{\sum_{i=1}^{N_t} o_{i,t}^{(p)} Rev_{i,t}} = \frac{\sum_{i=1}^{N_t} \frac{V_{i,t}^{(p)}}{Cap_{i,t}} E_{i,t}}{\sum_{i=1}^{N_t} \frac{V_{i,t}^{(p)}}{Cap_{i,t}} Rev_{i,t}} = \sum_{i=1}^{N_t} \frac{\alpha_{i,t}^{(p)}}{Cap_{i,t}} \frac{1}{\sum_{j=1}^{N_t} \frac{\alpha_{j,t}^{(p)}}{Cap_{j,t}} Rev_{j,t}} E_{i,t}$$

$$CI_t^{(p)} = \sum_{i=1}^{N_t} \alpha_{i,t}^{(p)} \frac{\frac{Rev_{i,t}}{Cap_{i,t}}}{\sum_{j=1}^{N_t} \alpha_{j,t}^{(p)} \frac{Rev_{j,t}}{Cap_{j,t}}} \frac{E_{i,t}}{Rev_{i,t}} = \sum_{i=1}^{N_t} \alpha_{i,t}^{(p)} \omega_{i,t}^{(p)} \frac{E_{i,t}}{Rev_{i,t}} = \sum_{i=1}^{N_t} \alpha_{i,t}^{(p)} \omega_{i,t}^{(p)} CI_{i,t}$$

- $\alpha_{i,t}^{(p)} = V_{i,t}^{(p)} / V_t^{(p)}$ is the weight of firm i in the portfolio ($V_t^{(p)} = \sum_{i=1}^{N_t} V_{i,t}^{(p)}$)
- $\omega_{i,t}^{(p)} = \frac{Rev_{i,t}}{Cap_{i,t}} / \sum_{j=1}^{N_t} \alpha_{j,t}^{(p)} \frac{Rev_{j,t}}{Cap_{j,t}}$

Remark: Only when all firms have the same revenue per market value, one has

$$CI_t^{(p)} = \sum_{i=1}^{N_t} \alpha_{i,t}^{(p)} CI_{i,t}$$

Carbon Intensity Measures

(2) Weighted-average carbon intensity: Portfolio's exposure to carbon-intensive companies, expressed in tCO₂e/mln \$:

$$WACI_t^{(p)} = \sum_{i=1}^{N_t} \alpha_{i,t}^{(p)} CI_{i,t} = \sum_{i=1}^{N_t} \alpha_{i,t}^{(p)} \frac{E_{i,t}}{Rev_{i,t}}$$

(3) Carbon footprint: Total carbon emissions for a portfolio normalized by the market value of the portfolio ("financed emissions"). This is the amount of annual carbon emissions that can be allocated to the investor per million dollars invested in the portfolio (tCO₂e/mln \$):

$$CF_t^{(p)} = \sum_{i=1}^{N_t} \alpha_{i,t}^{(p)} \frac{E_{i,t}}{Cap_{i,t}} = \frac{1}{V_t^{(p)}} \sum_{i=1}^{N_t} o_{i,t}^{(p)} E_{i,t}$$

Example of a Portfolio's Scope 1–3 Carbon Metrics (2019)

Market weights: based on market cap, Portfolio weights: my choice

	Walmart	Southern Company	Exxon Mobil	Portfolio
Scope 1–3 emissions (mln tonnes) ($E_{i,t}$)	59.8	96.6	227.3	–
Revenue (bln \$) ($Rev_{i,t}$)	514.4	20.7	255.6	–
Scope 1–3 intensity (tonnes/mln \$ rev)	116.3	4673.8	889.3	–
Market cap (bln \$) ($Cap_{i,t}$)	337.2	66.8	295.3	–
Relative market weight	0.482	0.096	0.422	1.0
Ptf market value (mln \$) ($V_{i,t}^{(p)}$)	31.9	12.5	55.6	100.0
Ptf weight ($\alpha_{i,t}^{(p)}$)	0.319	0.125	0.556	1.0
Ownership ($o_{i,t}^{(p)}$) (1/1000)	0.095	0.187	0.188	–
Ptf WA intensity (tonnes/mln \$ rev)	37.1	584.5	494.1	1115.8
Ptf intensity (tonnes/mln \$ rev)	–	–	–	660.7
Ptf footprint (tonnes/mln \$ invested)	56.6	180.8	427.7	665.2

Source: Trucost (2021).

Ownership:[5pt] The investor holds fraction 0.095/1000 of the market value of Walmart

$$37.1 = \alpha_{i,t} \times Cl_{i,t}$$

$$56.6 = o_{i,t} \times \frac{E_{i,t}}{V_t^{(p)}}$$

Avoided Emissions

Avoided emissions are emission reductions that occur outside of a product's life cycle or value chain, but because of the use of that product.

- Examples of products (goods and services) that avoid emissions
- low-temperature detergents
- fuel-saving tires
- teleconferencing services

Scope 1 and 2 emission reductions due to energy-saving and energy-mix process, do not qualify as “avoided emissions”

Emissions avoided by a company through the sale of low-carbon products / services lead to the reduction of the direct (Scope 1 and 2) emissions of the user or client

Avoided emissions contribute to the overall objective of reducing total emissions at the macroeconomic level

Difficult to include in a carbon footprint measure

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Exclusion/Best-in-Class Strategies

We assume an investor who does not want to invest in highly polluting firms. We define a highly polluting firm as a firm with a high carbon intensity.

- Define a threshold of the carbon intensity above which a given firm is excluded
- Define the threshold as corresponding to a given number of firms to exclude or, preferably, to a fraction of the market value of the portfolio that is excluded
- Decide what to do with the proceeds of the exclusion:
 - Reinvest proportionately in the remaining firms (**exclusion approach**)
 - Reinvest in the firms with the lowest carbon intensity in the same countries or sectors as excluded firms (**best-in-class approach with same region-sector exposures**)

Exclusion Strategy

Strategy rule:

1- Define the list of firms to be excluded from the BAU benchmark portfolio as the firms with the highest carbon intensity $Cl_{i,t}$:

$$I_{H,t} = \{1_{\{Cl_{i,t} > q_{\theta,t}\}}\}_{i=1}^{N_t}$$

where $q_{\theta,t}$ is the carbon intensity threshold corresponding to a probability θ of the carbon intensity distribution (say 99%). The targeted “exclusion probability” is $(1 - \theta)$.

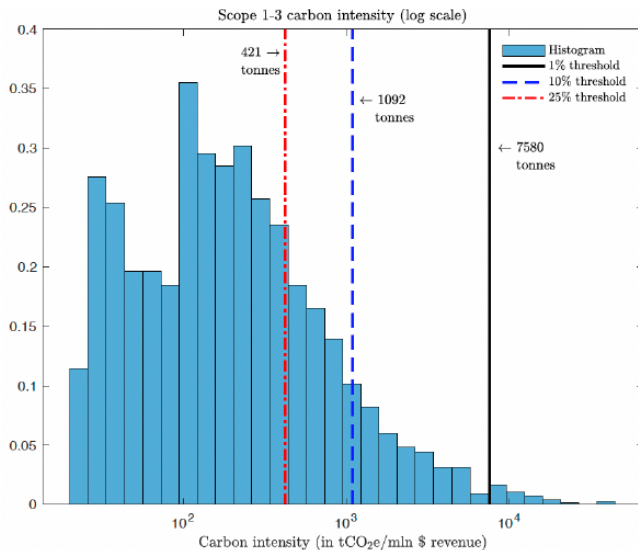
$q_{\theta,t}$ is defined such that the sum of the market weights of excluded firms, $\sum_{i=1}^{N_t} \alpha_{i,t}^{(b)} 1_{\{Cl_{i,t} > q_{\theta,t}\}}$, is as close as possible to the targeted probability, $(1 - \theta)$

2- Define the number of excluded firms as $N_{H,t} = \sum_{i=1}^{N_t} 1_{\{Cl_{i,t} > q_{\theta,t}\}}$

3- The proceeds, which represent a fraction $(1 - \theta)$ of the BAU benchmark market value, are reallocated proportionately to all stocks remaining in the portfolio

Exclusion Strategy: Logic

The exclusion or best-in-class strategy benefits from the **extreme asymmetry** in the distribution of carbon intensity



Exclusion Strategy: Weights

Vector of weights:

$$\alpha_{i,t}^{(p)} = 0 \quad \text{for } i \in I_{H,t} \quad \text{with } \sum_{i \in I_{H,t}} \alpha_{i,t}^{(b)} \approx 1 - \theta$$

$$\alpha_{i,t}^{(p)} = \alpha_{i,t}^{(b)} \left(\frac{1}{\sum_{i \in I_{l,t}} \alpha_{i,t}^{(b)}} \right) \quad \text{for } i \in I_{l,t}$$

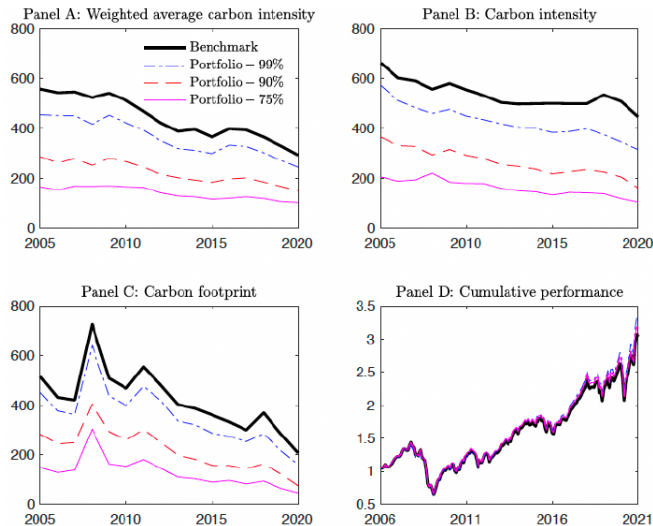
where $I_{l,t}$ is the list of firms included in the portfolio (complementary to $I_{H,t}$)

Very effective

... but impact on regional and sectoral exposures

- Firms from the utilities, energy, and materials are excluded
- Proceeds are reinvested into financials, healthcare or IT
- Severe underweighting of Emerging countries

Exclusion Strategy: Performance



Source: Jondeau, Mojon, and Pereira da Silva (2021). Intensity and footprint are in tonnes CO₂e/million

Excluded Firms across Sectors

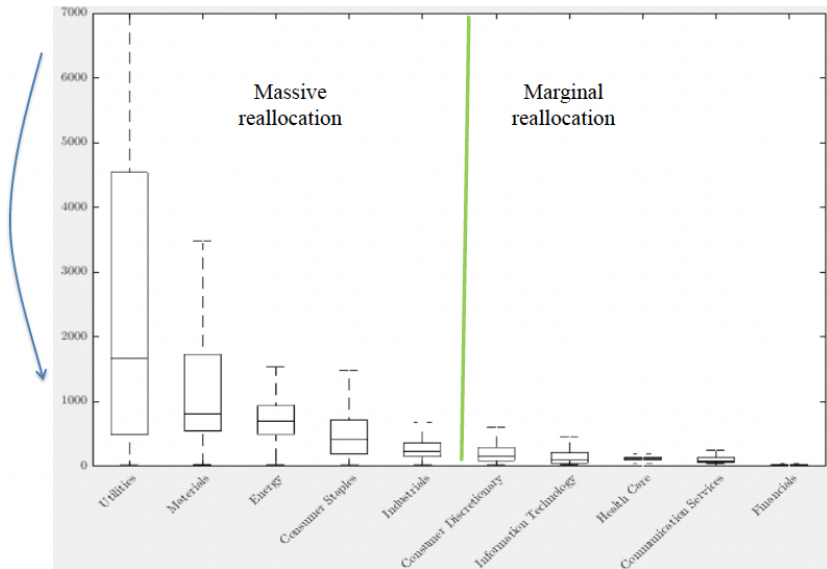
	Whole sample				2020			
	BAU bench.	Exclusion threshold 75%	90%	99%	BAU bench.	Exclusion threshold 75%	90%	99%
Financials	16.8	1.0	1.1	3.2	16.2	0.8	0.6	2.6
Industrials	16.3	15.8	13.2	2.5	14.3	17.9	9.3	–
Consumer discretionary	14.4	5.8	1.6	0.2	13.7	10.4	2.6	–
Materials	9.9	31.4	36.4	36.9	9.2	25.3	33.8	41.0
Information technology	8.8	2.2	0.7	0.1	11.6	4.8	2.0	1.3
Consumer staples	7.9	13.4	10.3	0.1	8.6	15.3	15.7	–
Health care	6.1	0.6	0.1	–	9.3	1.7	1.0	–
Utilities	5.6	15.2	24.0	53.8	5.0	12.1	18.1	50.0
Real estate	5.5	0.7	0.5	–	5.8	1.2	0.6	–
Energy	5.3	13.8	12.0	3.3	3.7	10.0	16.3	5.1
Communication services	3.4	0.1	0.0	–	2.6	0.6	–	–

Excluded Firms across Countries

	Whole sample				2020			
	BAU bench.	Exclusion 75%	threshold 90%	99%	BAU bench.	Exclusion 75%	threshold 90%	99%
United States	21.0	17.5	16.2	19.2	19.6	15.7	15.9	16.7
Japan	14.2	13.3	9.6	3.4	10.4	10.7	6.8	1.3
China	6.7	7.7	8.7	15.7	21.8	21.8	19.9	24.4
Taiwan	4.2	4.4	4.2	3.2	2.9	3.7	2.6	1.3
United Kingdom	4.1	2.8	2.2	0.5	3.1	2.2	2.4	1.3
Korea	4.1	3.8	3.1	1.3	3.6	3.4	2.0	–
Canada	3.5	4.5	4.2	2.5	2.9	3.5	4.8	1.3
Hong Kong	3.5	3.3	4.4	9.2	3.1	3.9	4.2	7.7
India	2.9	3.8	4.5	14.5	3.2	4.5	4.2	10.3
France	2.9	2.0	2.1	–	2.4	1.7	1.4	–

Source: Jondeau, Mojon, and Pereira da Silva (2021). Intensity and footprint are in tonnes CO₂e/million \$.

Best-In-class Strategy: In practice



Best-In-class Strategy

Strategy rule:

1- Denote by R_i and S_i the region and sector of firm i . The indicator variable $1_{\{R_i=r, S_i=s\}}$ is equal to 1 if the firm belongs to region r and sector s and 0 otherwise.

2- Denote the set of firms in region r and sector s by $I_t(r, s) = \{1_{R_i=r, S_i=s}\}_{i=1}^{N_t}$. The list of firms to be excluded ($I_{H,t}$) in region r and sector s is

$$I_{H,t}(r, s) = \left\{ 1_{\{R_i=r, S_i=s, Cl_{i,t} > q_{\theta,t}\}} \right\}_{i=1}^{N_t}$$

A proportion $\theta_t(r, s) = \sum_{i \in I_{H,t}(r, s)} \alpha_{i,t}^{(b)} / \sum_{i \in I_t(r, s)} \alpha_{i,t}^{(b)}$ of the market value is excluded.

3- The proceeds are reinvested in the same region-sector in the set of firms with the lowest carbon intensity with a cumulative market value approximately the same. The set of overweighted firms is defined as $I_{L,t}(r, s) = \left\{ 1_{\{R_i=r, S_i=s, Cl_{i,t} \leq q_{\theta_t(r, s), t}\}} \right\}_{i=1}^{N_t}$.

The overall set of overweighted firms is given by $I_{L,t} = \{I_{L,t}(r, s)\}_{r,s}$.

Best-In-class Strategy

Vector of weights:

$$\alpha_{i,t}^{(p)} = 0 \quad \text{for } i \in I_{H,t} \quad \text{with} \quad \sum_{i \in I_{H,t}} \alpha_{i,t}^{(b)} \approx 1 - \theta$$

$$\alpha_{i,t}^{(p)} = \alpha_{i,t}^{(b)} \quad \text{for } i \in I_{L,t}$$

$$\alpha_{i,t}^{(p)} = \alpha_{i,t}^{(b)} \left(1 + \frac{\sum_{j \in I_{H,t}(R_i, S_i)} \alpha_{i,t}^{(b)}}{\sum_{j \in I_{L,t}(R_i, S_i)} \alpha_{i,t}^{(b)}} \right) \quad \text{for } i \in I_{L,t}$$

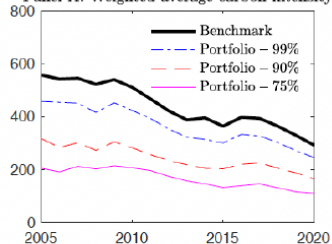
where $\sum_{j \in I_{L,t}(R_i, S_i)} \alpha_{i,t}^{(b)} \approx \sum_{j \in I_{H,t}(R_i, S_i)} \alpha_{i,t}^{(b)}$

Very effective

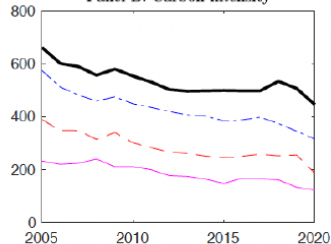
... with the same region-sector exposures as the benchmark

Best-In-class Strategy

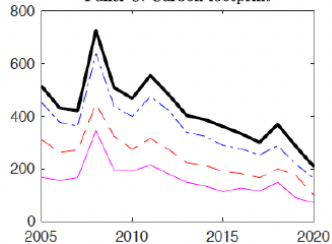
Panel A: Weighted average carbon intensity



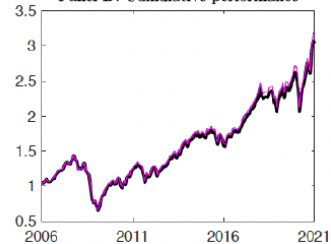
Panel B: Carbon intensity



Panel C: Carbon footprint



Panel D: Cumulative performance



Exclusion/Best-in-Class Strategies

Reduction of portfolio carbon emissions (2005-2020) – Scope 1+2+3 upstream

	All firms	Exclusion approach			Best-in-class approach		
		99%	90%	75%	99%	90%	75%
Weighted avg carbon intensity	440	362.5	222.8	139.9	364.2	243.2	168.1
Reduction (in %)	(–)	(17.6)	(49.3)	(68.0)	(17.2)	(44.6)	(61.9)
Carbon footprint	535	425.6	263.3	161.6	426.6	284.7	184.9
Reduction (in %)	(–)	(20.7)	(51.2)	(70.0)	(20.6)	(47.2)	(65.7)
Carbon intensity	422.2	356.2	218.0	128.4	356.9	245.4	160.2
Reduction (in %)	(–)	(16.4)	(49.7)	(70.7)	(16.2)	(42.6)	(62.8)
Annual return (in %)	8.67	8.69	8.93	9.15	8.68	8.69	8.69
Annual volatility (in %)	16.3	16.3	16.1	16.4	16.2	16.1	16.3
Sharpe ratio	0.53	0.53	0.55	0.56	0.53	0.54	0.53
Tracking error (in %)	(–)	0.09	0.71	1.71	0.07	0.44	1.06

Source: Jondeau, Mojon, and Pereira da Silva (2021). Intensity and footprint are in tonnes CO₂e/million \$.

Minimum Variance Portfolio

We consider the Global Minimum Variance (GMV) Portfolio, which solves the program

$$\begin{aligned} \min_{\{\alpha\}} \quad & \frac{1}{2} \alpha_t' \Sigma_t \alpha_t \\ \text{s.t.} \quad & \alpha_t' \mathbf{e} = 1 \end{aligned}$$

The solution is:

$$\alpha_t^* = \frac{\Sigma_t^{-1} \mathbf{e}}{\mathbf{e}' \Sigma_t^{-1} \mathbf{e}}$$

Portfolio Optimization with Carbon Risk – Adding Restrictions

Fundamental-based risk management: We restrict the exposure of the portfolio to carbon intensity (idiosyncratic carbon risk)

- Individual threshold: $\alpha_{i,t} = 0$ if $CI_{i,t} \geq CI_t^+$
- Portfolio threshold: $CI(\alpha_t) = \sum_{i=1}^N \alpha_{i,t} CI_{i,t} \leq CI_t^+$

where $CI(\alpha_t)$ is the weighted average carbon intensity of the portfolio

The program to solve is:

$$\begin{aligned} \min_{\{\alpha\}} \quad & \frac{1}{2} \alpha_t' \Sigma_t \alpha_t \\ \text{s.t.} \quad & \begin{cases} \alpha_t' \mathbf{e} = 1 \\ CI(\alpha_t) \leq CI_t^+ \end{cases} \end{aligned}$$

End of Lecture 7